



Week 7 Workshop Evaluation

IFN647

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Workshop Objectives

- Develop a clear understanding of the fundamental evaluation metrics in Information Retrieval (IR), including **Precision**, **Recall**, and the **F1-measure**, as well as rank-based metrics such as **Precision@n** and **Recall@n** for evaluating ranked retrieval results.
- Gain the ability to systematically use **relevance judgements** (ground-truth labels indicating whether documents are relevant to a query) to assess the effectiveness of IR models.
- Build practical skills by implementing Python programs that:
 - Compute **Precision**, **Recall**, and **F1-measure** based on model outputs and relevance judgements.
 - Evaluate ranked retrieval results by calculating **Precision@k** and **Recall@k** at various cutoff positions.
 - Identify rank positions where relevant documents are retrieved, compute Precision at those positions, and use these values to calculate **Average Precision (AP)** for a query.

Workshop Task 1.1

Task 1. Understand the definitions of Precision, recall and F measure and discuss how to use the following definitions. You may discuss these measures with your classmate or your tutor if you have any doubts.

(1) Assume A is a set of relevant documents (e.g., a benchmark, or relevance judgments), B is a set of retrieved documents (e.g., the output of an IR model, a binary output)

- **binary_output.txt**

```
R105 2493 1
R105 2494 0
R105 3008 1
R105 5004 0
R105 5223 0
R105 5225 1
R105 5226 1
R105 6635 0
R105 7937 0
R105 15744 1
R105 17650 0
R105 22961 0
R105 25263 0
R105 26711 0
R105 27966 0
R105 28198 0
```

- **relevance_judgment.txt**

```
R105 2493 1
R105 2494 1
R105 3008 1
R105 5004 1
R105 5223 0
R105 5225 1
R105 5226 1
R105 6635 0
R105 7937 0
R105 15744 1
R105 17650 0
R105 22961 0
R105 25263 0
R105 26711 0
R105 27966 0
R105 28198 0
```

Check rel_data Folder for data files

Format:

***Topic* DocumentID Relevance**

Workshop Task 1.1

Task 1. Understand the definitions of Precision, recall and F measure and discuss how to use the following definitions. You may discuss these measures with your classmate or your tutor if you have any doubts.

- (1) Assume A is a set of relevant documents (e.g., a benchmark, or relevance judgments), B is a set of retrieved documents (e.g., the output of an IR model, a binary output)

	A (Relevant)	$\bar{A}$ (Non-Relevant)
B (Retrieved)	(1) $A \cap B$ (✓ TP)	(3) $\bar{A} \cap B$ (✗ FP)
$\bar{B}$ (Not Ret.)	(2) $A \cap \bar{B}$ (✗ FN)	(4) $\bar{A} \cap \bar{B}$ (✓ TN)

$$Recall = \frac{|A \cap B|}{|A|}$$

$$Precision = \frac{|A \cap B|}{|B|}$$

F-Measure (F1),

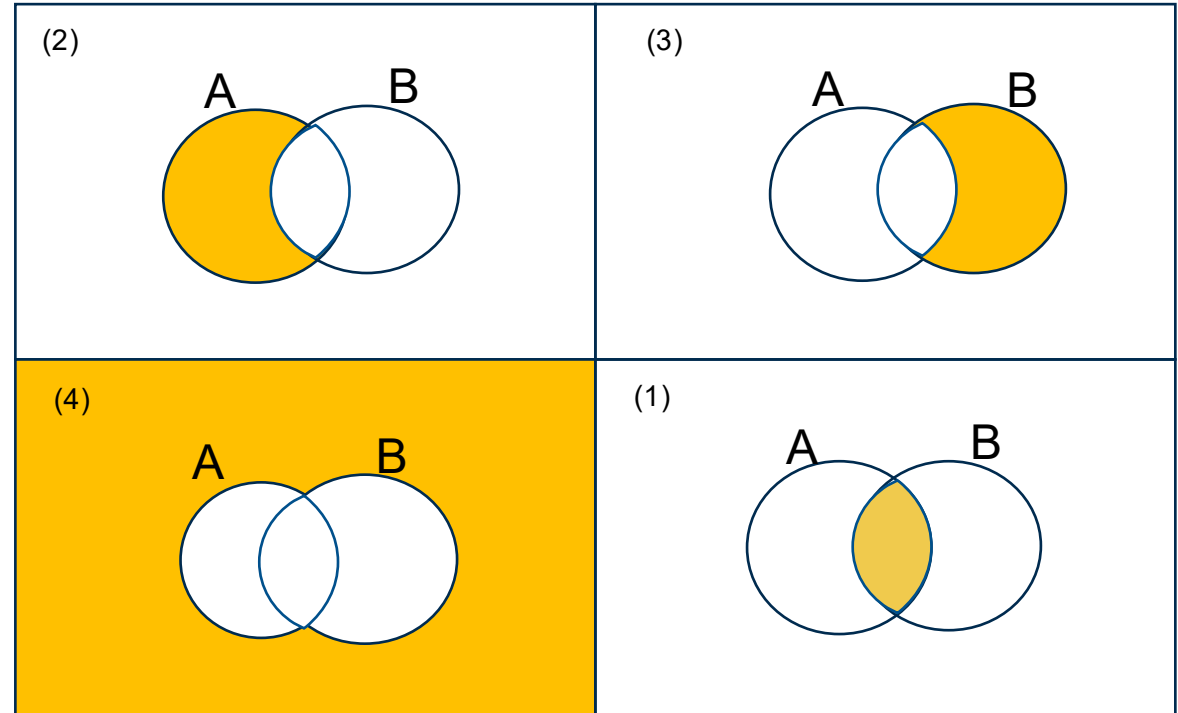
$$F = \frac{1}{\frac{1}{2}(\frac{1}{R} + \frac{1}{P})} = \frac{2RP}{(R+P)}$$

A is a set of relevant documents

$\bar{A}$ is a set of non-relevant documents

B is a set of retrieved documents

$\bar{B}$ is a set of non-retrieved documents



Workshop Task 1.2

- ranked_output.txt

(2) Assume an IR model returns a list of ranked documents (in descending order by the document weight or score). At each position n , we have the following definition of precision (recall) at position n :

$$Recall@n = \frac{|A \cap B_n|}{|A|}$$

Of all the relevant documents, how many did we retrieve in the top n ?

$$Precision@n = \frac{|A \cap B_n|}{n}$$

Of the top n documents the model returned, how many were actually relevant?

where B_n is the set of **top- n** documents in the output of the IR model.

```
R105 2493 1.0
R105 3008 0.9
R105 5225 0.8
R105 5226 0.7
R105 15744 0.6
R105 31141 0.6
R105 40239 0.5
R105 48148 0.5
R105 49633 0.4
R105 51493 0.4
R105 59813 0.3
R105 65210 0.3
R105 80484 0.3
R105 86961 0.3
R105 2494 0.2
R105 5004 0.19
R105 5223 0.18
R105 6635 0.17
R105 7937 0.16
R105 17650 0.15
R105 22961 0.14
R105 25263 0.13
```

These definitions are useful (e.g., for calculating average precisions) for IR models that rank documents

Workshop Task 2

INPUT

• **relevance_judgment.txt**

```
R105 2493 1
R105 2494 1
R105 3008 1
R105 5004 1
R105 5223 0
R105 5225 1
R105 5226 1
R105 6635 0
R105 7937 0
R105 15744 1
R105 17650 0
R105 22961 0
R105 25263 0
R105 26711 0
R105 27966 0
R105 28198 0
R105 31141 0
R105 33961 0
R105 38229 0
R105 40239 1
R105 40259 1
...
```

• **binary_output.txt**

```
R105 2493 1
R105 2494 0
R105 3008 1
R105 5004 0
R105 5223 0
R105 5225 1
R105 5226 1
R105 6635 0
R105 7937 0
R105 15744 1
R105 17650 0
R105 22961 0
R105 25263 0
R105 26711 0
R105 27966 0
R105 28198 0
```

OUTPUT

```
Rel_doc = {'2493': 1, '2494': 1, '3008': 1, '5004': 1,
'5223': 0, '5225': 1, '5226': 1, '6635': 0, '7937': 0,
'15744': 1, '17650': 0, '22961': 0, '25263': 0,
'26711': 0, '27966': 0, '28198': 0, '31141': 0,
'33961': 0, ...}
```

```
Ret_doc = {'2493': 1, '2494': 0, '3008': 1, '5004': 0,
'5223': 0, '5225': 1, '5226': 1, '6635': 0, '7937': 0,
'15744': 1, '17650': 0, '22961': 0, '25263': 0, ...}
```

```
The number of relevant documents: 16
```

```
The number of retrieved documents: 14
```

```
The number of retrieved documents that are relevant: 11
```

```
recall = 0.6875
```

```
precision = 0.7857142857142857
```

```
F-Measure = 0.7333333333333334
```

Task 2: read a topic-doc-assignment file (e.g., *relevance_judgments.txt*, the benchmark) and a retrieved topic-doc-assignment file (e.g., *binary_output.txt*, the output of an IR model for query *R105*); and calculate the IR model's Recall, Precision, and F-Measure (F1).

- Please download two topic-doc-assignment files (*rel_data.zip*) and save them in a folder (e.g., “rel_data”). Both files have the format of

Topic DocumentID Relevance

For the file “relevance_judgments.txt”, *Relevance* (Relevance Judgment) = “1” indicates relevant and “0” means non-relevant. For the file “binary_output.txt”, the *relevance* values are generated by the IR model. We can obtain a set of retrieved documents by selecting the rows with *Relevance* (Relevance Values) = “1”.

- Define a function *rel_setting(inputpath)*, which reads the two topic-doc-assignment files in the folder *inputpath*, and returns a pair of dictionaries {*documentID: Relevance_judgment, ...*} and {*documentID: Relevance_value, ...*} for all documents in “relevance_judgments.txt”, and “binary_output.txt”, respectively.
- Define a main function to call function *rel_setting()*, calculate Recall, Precision, and F-measure and display the result.

Workshop Task 3

Task 3: Read the ranked output file (“ranked_output.txt”, the ranked output of another IR model for query *R105*), and calculate its average precision.

- Please download the ranked output file and save it in the same folder you created for Task 2.
- Extend function `rel_setting(inputpath)` to return 3 dictionaries (two for task 2 and one for task 3). The dictionary for task 3 should have the format of `{rankingNo: documentID, ...}`, and it only includes top-10 documents (ranked in descending order), where $rankingNo = 1, 2, \dots, 10$.
- Extend the main function to calculate Recall and Precision at the rank positions where a relevant document was retrieved, and then calculate the average precision, and print out the result.

- ranked_output.txt

```
R105 2493 1.0
R105 3008 0.9
R105 5225 0.8
R105 5226 0.7
R105 15744 0.6
R105 31141 0.6
R105 40239 0.5
R105 48148 0.5
R105 49633 0.4
R105 51493 0.4
R105 59813 0.3
R105 65210 0.3
R105 80484 0.3
R105 86961 0.3
R105 2494 0.2
R105 5004 0.19
R105 5223 0.18
R105 6635 0.17
R105 7937 0.16
R105 17650 0.15
R105 22961 0.14
R105 25263 0.13
```


Workshop Task 3

```
C = {'2493': 1.0, '3008': 0.9, '5225': 0.8, '5226': 0.7, '15744': 0.6, '31141': 0.6,
      '40239': 0.5, '48148': 0.5, '49633': 0.4, '51493': 0.4, '59813': 0.3, '65210': 0.3,...
```

Example of output

```
R = {1: '2493', 2: '3008', 3: '5225', 4: '5226', 5: '15744', 6: '31141', 7:
      '40239', 8: '48148', 9: '49633', 10: '51493'}
```

```
At position 1docID: 2493, precision= 1.0
At position 2docID: 3008, precision= 1.0
At position 3docID: 5225, precision= 1.0
At position 4docID: 5226, precision= 1.0
At position 5docID: 15744, precision= 1.0
At position 7docID: 40239, precision= 0.8571428571428571
At position 8docID: 48148, precision= 0.875
At position 9docID: 49633, precision= 0.8888888888888888
At position 10docID: 51493, precision= 0.9
The average precision = 0.9467813051146384
```

- ranked_output.txt

```
R105 2493 1.0
R105 3008 0.9
R105 5225 0.8
R105 5226 0.7
R105 15744 0.6
R105 31141 0.6
R105 40239 0.5
R105 48148 0.5
R105 49633 0.4
R105 51493 0.4
R105 59813 0.3
R105 65210 0.3
R105 80484 0.3
R105 86961 0.3
R105 2494 0.2
R105 5004 0.19
R105 5223 0.18
R105 6635 0.17
R105 7937 0.16
R105 17650 0.15
R105 22961 0.14
R105 25263 0.13
```


Sample Task3 Results

For task 3:

```
At position 1docID: 2493, precision= 1.0
At position 2docID: 3008, precision= 1.0
At position 3docID: 5225, precision= 1.0
At position 4docID: 5226, precision= 1.0
At position 5docID: 15744, precision= 1.0
At position 7docID: 40239, precision= 0.8571428571428571
At position 8docID: 48148, precision= 0.875
At position 9docID: 49633, precision= 0.8888888888888888
At position 10docID: 51493, precision= 0.9
The average precision = 0.9467813051146384
```

Of the top 10 documents the model returned, on average 95% are actually relevant.